

Kidney Diseases Associated With Coronaviruses

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INTRODUCTION

The first cases of COVID-19 were reported in Wuhan, China, in late 2019, and by mid-March 2020, the World Health Organization officially declared a pandemic.¹ COVID-19 has since caused millions of deaths worldwide, and acute kidney injury (AKI) and other kidney syndromes are important complications COVID-19.² Moreover, COVID-19 has disproportionately impacted persons with kidney failure who receive chronic dialysis and are kidney transplant recipients (KTRs).

SARS-COV-2 AND VIRAL PATHOGENESIS

The virus that causes COVID-19 is SARS-CoV-2. Coronaviruses are enveloped, single-stranded RNA viruses that infect humans, other mammals, and nonmammalian species.¹ The first coronavirus found to induce severe disease in humans was severe acute respiratory syndrome coronavirus (SARS-CoV), which caused the first reported SARS outbreak in China in 2002 and subsequent sporadic outbreaks. In 2012, the Middle Eastern respiratory syndrome (MERS), caused by MERS-CoV, was reported on the Arabian Peninsula. SARS-CoV and MERS-CoV use the same receptor (ACE2) as SARS-CoV-2 to infect host cells.³ Kidney injury is a common complication of MERS but probably occurs less often in patients with SARS.^{4,5}

Cellular entry of SARS-CoV-2 is mediated by binding of the viral spike (S) protein to ACE2 on the surface of target cells and subsequent cleavage/activation of S by the type 2 transmembrane serine protease (TMPRSS2), thereby enabling the S protein to mediate entry of the virus into the host cell (Fig. 59.1), leading to cellular injury and activation of inflammatory responses.¹

Robust innate immune responses, including interferon production early after SARS-CoV-2 infection, are important for limiting the severity of COVID-19.⁶ However, in hospitalized patients with established COVID-19, high levels of inflammatory mediators predict increased risk of AKI, mechanical ventilation, and mortality, whereas immunosuppression with high-dose corticosteroids reduces mortality.^{1,7} Similarly, dysregulated activation of coagulation in COVID-19 causes microvascular and, less often, macrovascular occlusion, which is an important contributor to lung injury,¹ but the role of dysregulated coagulation in COVID-19-associated kidney injury is less clear.

DOES SARS-COV-2 INFECT THE KIDNEY?

The receptor ACE2 is strongly expressed in proximal tubular cells, whereas other kidney cells, including podocytes, express ACE2 at lower levels.⁸ Studies have reported the presence of viral antigens and nucleic acids in kidney autopsy and biopsy specimens, and electron

micrographs of intracellular structures that appeared to be consistent with SARS-CoV-2 virions.⁸ Subsequent studies suggested that technical limitations of the assays used for SARS-CoV-2 antigen and nucleic acid detection as well as difficulty in differentiating intracellular organelles from virions may have led to some false positive results.⁹ More recently, viral RNA and protein have been detected in kidney tubular cells, where the nucleocapsid and spike proteins colocalized with ACE2.¹⁰ Therefore, it is likely that kidney tubular cells become infected by SARS-CoV-2, but it is unknown if this causes clinically apparent kidney injury.

COVID-19 ACUTE KIDNEY INJURY

Epidemiology

The incidence of COVID-19-associated AKI (COVID-19 AKI) is high, ranging from 17% to 46% in large cohorts of hospitalized patients with COVID-19.¹¹⁻¹³ This variability likely reflects differences in disease severity, age, and comorbidities, and it is not clear whether the incidence and severity of AKI is higher in patients with COVID-19 than otherwise similar patients without COVID-19. There is significant heterogeneity in the severity of COVID-19 AKI with the proportion of stage 3 AKI, ranging from 11% to 42% with as many as 19% requiring kidney replacement therapy (KRT).¹²⁻¹⁴ In a multicenter cohort study of intensive care unit (ICU) patients from 67 US hospitals, 21% developed AKI requiring KRT within 14 days of ICU admission.⁷

Risk Factors for COVID-19 AKI

Although many factors contribute to AKI in patients with COVID-19 (Fig. 59.1), hypovolemia and sepsis are the most common causes reported.¹³ Risk factors associated with the incidence and severity of COVID-19 AKI include older age, male sex, Black race, obesity, hypertension, and diabetes mellitus. The presence of chronic kidney disease (CKD) and CKD stage are strongly associated with increased risk of COVID-19 AKI and the need for KRT.^{7,11-13}

Increased severity of illness, including hypotension, hypoxia, and the need for mechanical ventilation are associated with increased risk of COVID-19 AKI.^{7,13} Hypovolemia is a major contributor to COVID-19 AKI. COVID-19 illness leads to hypovolemia by increasing insensible losses (fever and tachypnea), reducing oral intake (sore throat, nausea, and anorexia), and increasing gastrointestinal losses (diarrhea).¹⁵ Some groups have reported partial Fanconi syndrome with wasting of low-molecular-weight proteins, amino acids, uric acid, and phosphate but not glucose.¹⁵ It is unknown whether primary tubular wasting contributes to hypovolemia in COVID-19.

Elevated serum biomarkers of inflammation (C-reactive protein and ferritin) and/or coagulation (D-dimer) are associated with

Pathogenesis of Acute Kidney Injury in COVID-19

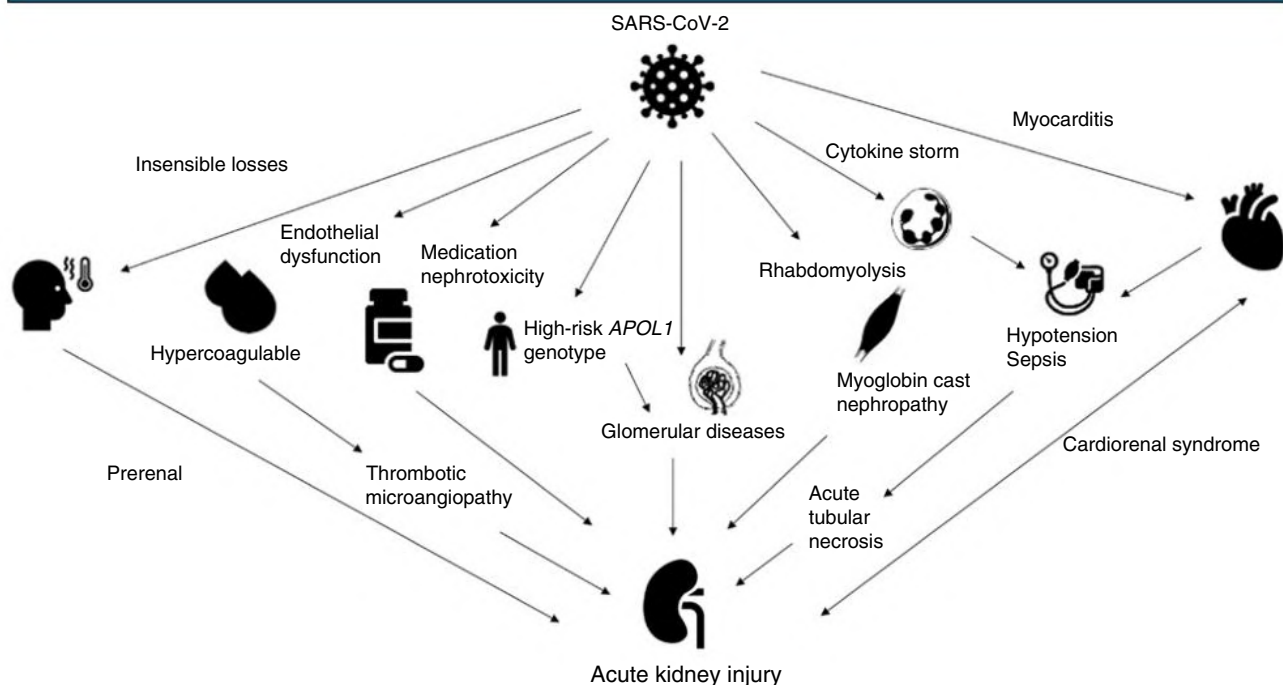


Fig. 59.1 Pathogenesis of acute kidney injury in patients with COVID-19.

increased risk of COVID-19 AKI in retrospective analyses⁷; however, the value of biomarkers as independent predictors of COVID-19 AKI remains to be demonstrated.

Histopathology of COVID-19 AKI

Small case series have reported pathologic findings in autopsy and kidney biopsy specimens from patients with COVID-19 AKI. Acute tubular injury (ATI) or tubular necrosis were identified in up to 90%, and glomerular and peritubular capillary congestion were common findings without additional glomerular lesions or fibrin thrombi¹⁶⁻²⁰ (Fig. 59.2). Myoglobin casts indicative of rhabdomyolysis have been found in up to 10% of biopsies showing ATI.

Treatment of COVID-19 AKI

There are no specific treatments to prevent or reduce the severity or duration of COVID-19 AKI. Treatment is supportive and includes optimization of kidney perfusion and minimizing exposures that may exacerbate kidney injury. Though studies have not demonstrated that a specific KRT modality results in superior clinical outcomes in patients with COVID-19, continuous kidney replacement therapy (CKRT) is preferred in hemodynamically unstable patients. Because patients with COVID-19 often require prone positioning to improve oxygenation, placement of dialysis catheters in the internal jugular rather than the femoral vein allows for easier access during proning. Anticoagulation is often required during KRT because patients with COVID-19 are at increased risk of thrombotic complications, including vascular access and dialysis circuit clotting. Small retrospective studies have reported increased CKRT filter life span with anticoagulation, but the optimal approach to anticoagulation for CKRT in patients with COVID-19 remains undetermined.²¹

In regions where the incidence of COVID-19 AKI has overwhelmed KRT capacity, several strategies have been used to increase KRT availability. Increased blood and dialysate flow rates and shortened

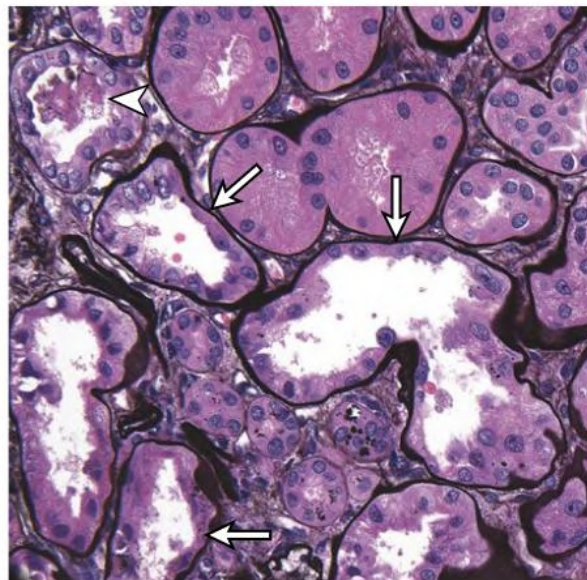


Fig. 59.2 Acute Tubular Injury in a Patient With COVID-19 and Acute Kidney Injury. Proximal tubular cells show loss of brush border staining, flattening, and degenerative change (arrows) with cellular debris in a tubular lumen (arrowhead).

treatments for intermittent HD can allow for a greater number of treatments. Similarly, flows on CKRT machines can be increased to provide prolonged intermittent kidney replacement therapy, allowing CKRT machines to be used for multiple patients each day.²¹ Finally, several centers have reported using acute peritoneal dialysis in patients with COVID-19 AKI.²¹ Important advantages of peritoneal dialysis include scalability due to lack of requirement for a dialysis machine or high volumes of dialysate-quality water, and elimination of issues with dialysis circuit and vascular access clotting. Though studies reported

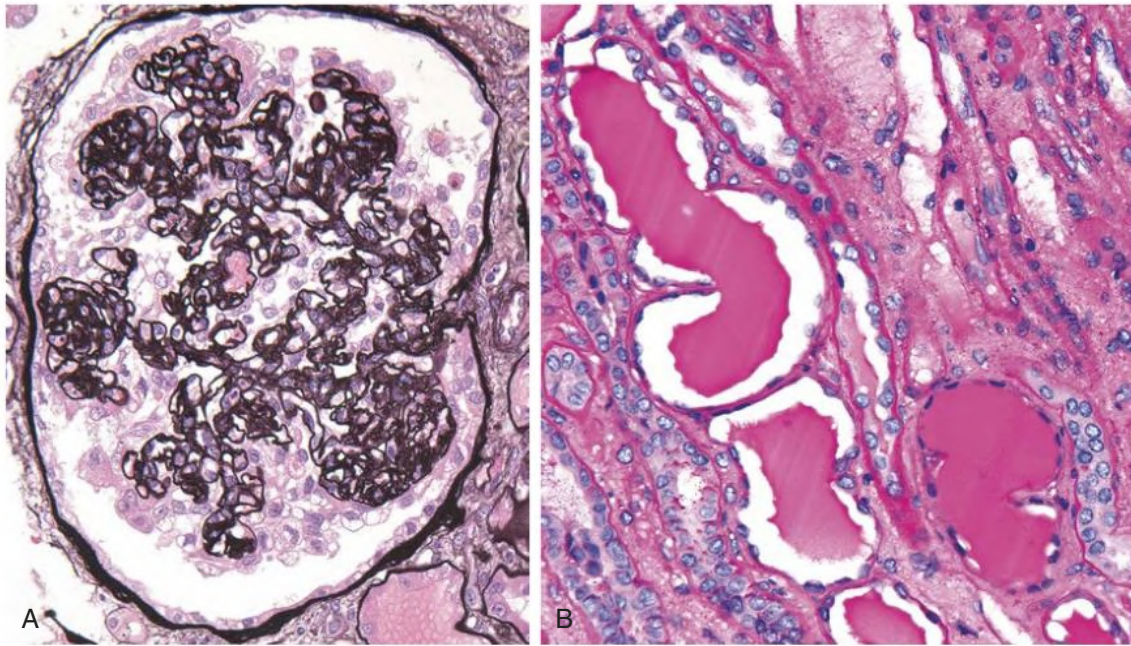


Fig. 59.3 Collapsing Glomerulopathy in an African American Patient With Moderately Symptomatic COVID-19 and Nephrotic Syndrome. (A) Global epithelial cell hypertrophy and hyperplasia with wrinkling and retraction of glomerular capillaries. (B) Microcystic tubules with proteinaceous casts.

clinical outcomes that appear to be acceptable with acute peritoneal dialysis in patients with COVID-19 AKI, peritoneal dialysis was sometimes difficult to perform in the most acutely ill patients, especially those who required prone positioning.

Prognosis of COVID-19 AKI

COVID-19 AKI often has a prolonged course and 35% to 40% of hospitalized patients with COVID-19 AKI meet criteria for acute kidney disease (i.e., AKI persisting for >7 days).^{12,13} Of those with acute kidney disease, 36% had resolution to baseline kidney function during follow-up. However, 14% of patients with complete AKI resolution at discharge subsequently developed acute kidney disease, and 30% of survivors with AKI requiring KRT remained dialysis-dependent at the time of hospital discharge.¹²

Mortality among patients with COVID-19 AKI is high, ranging from 19% in those with stage 1 AKI to more than 60% in those with stage 3D AKI and/or highest severity of illness.^{7,11-13} Among ICU patients with COVID-19 AKI, older age, receipt of two or more vasopressors at time of KRT initiation, and daily urine output less than 100 mL are associated with higher 28-day mortality.⁷

COVID-19–ASSOCIATED GLOMERULAR DISEASES

Clinical Presentation and Histopathology

Collapsing glomerulopathy (CG), minimal change disease (MCD), membranous nephropathy, and pauci-immune crescentic glomerulonephritis are the predominant glomerular lesions observed in patients with COVID-19.^{22-24a} Clinical features of COVID-19–associated CG and MCD are new onset nephrotic syndrome or nephrotic range proteinuria with AKI. Collapsing glomerulopathy (Fig. 59.3) has been reported predominantly in persons of African ancestry, and high-risk *APOL1* genotypes have been identified in the great majority of patients with CG who underwent testing, suggesting that genetic susceptibility likely contributes to risk of CG in patients with COVID-19.^{22-24a} Severe COVID-19 is associated with coagulopathy and microvascular thromboses which can resemble immune-triggered, complement-mediated

thrombotic microangiopathy (TMA). COVID-19–associated TMA (Fig. 59.4) is characterized by microangiopathic hemolytic anemia, thrombocytopenia, and severe AKI requiring KRT. Those with a predisposing risk factor for endothelial injury, such as hypertension, drug exposure, or a genetic or acquired defect in the alternative complement pathway, may be at increased risk.^{19,22,25} COVID-19 also has been associated with immune-mediated glomerular disease. Cases of anti-glomerular basement membrane (anti-GBM) disease, antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis, immunoglobulin A (IgA) nephropathy, and membranous nephropathy (MN) have been reported in patients with COVID-19.^{22,26,27} These glomerular diseases may manifest concurrently or several weeks after resolution of the acute infection, and it is unknown if this is chance occurrence or is pathogenetically linked. Interestingly, IgA nephropathy may occur de novo or be exacerbated after SARS-CoV-2 infection, and gross hematuria and/or transient worsening of kidney function have occurred after COVID-19 vaccination in patients with IgA nephropathy, possibly due to augmented immune injury.²⁸⁻³¹

Treatment and Prognosis

Despite a 50% or higher incidence of severe AKI requiring KRT in patients with COVID-19–associated CG, kidney function and proteinuria improve in many patients up to 90 days after resolution of infection. Therefore, those who are discharged on KRT should be monitored closely for recovery of kidney function.^{19,22,23,25} All patients should receive supportive care directed at blood pressure control and maintaining euolemia. Treatment with immunosuppression can be considered in those with severe nephrotic syndrome or persistent nephrotic range proteinuria that fails to improve despite resolution of infection, though potential risks of immunosuppression in patients with active or recent infection must be considered.

COVID-19–associated TMA is associated with a poor prognosis including dialysis dependence and high mortality. In the absence of controlled clinical studies to guide treatment, clinicians may consider interventions used for non-COVID-19–related TMA including plasma exchange and terminal complement inhibitors.

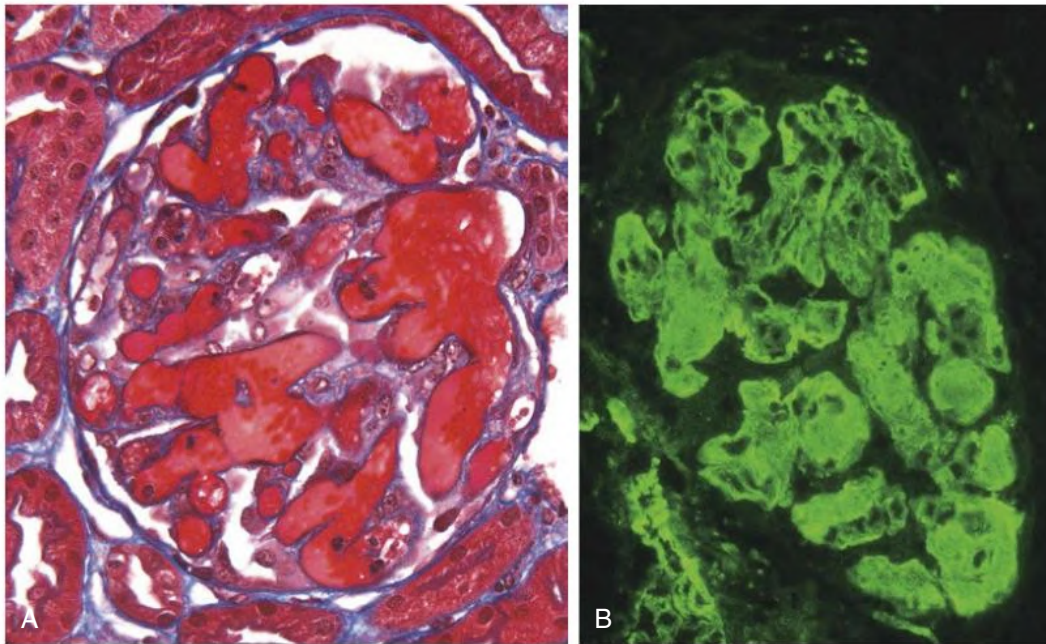


Fig. 59.4 Thrombotic Microangiopathy in a Patient With COVID-19 Presenting With Hypertension, Anemia, Thrombocytopenia, Hematuria, and Proteinuria. (A) Glomerulus with global capillary thrombosis. (B) Immunofluorescence showing extensive glomerular fibrin deposition.

COVID-19 IN PATIENTS WITH KIDNEY FAILURE REQUIRING CHRONIC DIALYSIS

COVID-19 incidence and mortality are significantly increased in patients with end-stage kidney disease (ESKD). Male sex, older age, and residence in a congregate setting have been consistently identified as risk factors for contracting COVID-19, whereas home dialysis has been associated with a reduced risk of infection compared with in-center dialysis.³²

Hospitalization rates approach 60% or higher among chronic dialysis patients with COVID-19 compared with 15% to 20% observed in the general population.^{33,34} Moreover, ESKD patients hospitalized with COVID-19 have 1.4-fold higher odds of in-hospital death after adjusting for demographics and comorbidities, suggesting ESKD is an independent risk factor for mortality due to COVID-19.³⁵ National and international registries estimate short-term mortality of 20% or greater among all dialysis patients with COVID-19 and 40% mortality in those who require hospitalization, underscoring the importance of prevention in this vulnerable population.³⁶

Patients receiving in-center dialysis are at increased risk for contracting COVID-19 due to the requirement of transportation to and from dialysis and frequent contact with caregivers, health care providers, nonmedical staff, and other patients.³⁷ Thus, prevention of infection at the dialysis unit is critical. Screening for signs and symptoms of COVID-19 should be performed prior to each dialysis session. Additionally, some centers perform weekly point-of-care testing of all dialysis patients, which may lead to early detection of COVID-19 prior to symptom onset and may help prevent disease spread. For patients with suspected or confirmed COVID-19, dialysis must be provided in a location and manner that minimizes risk to staff and patients.³⁷

COVID-19 IN KIDNEY TRANSPLANT RECIPIENTS

Epidemiology and Risk Factors

Compared to the general population, kidney transplant recipients are at increased risk of hospitalization and death from COVID-19 due to

treatment with immunosuppressive medications and excess burden of comorbidities. The incidence of AKI may exceed 50%, and allograft loss has been reported in 4% to 11% of patients.³⁸ Large registries of KTRs with symptomatic COVID-19 report hospitalization rates of 87% to 100% and mortality of 19% to 32%.^{38,39} However, routine testing for SARS-CoV-2 IgG antibodies in asymptomatic patients suggests that a significant proportion of KTRs have a mild disease course. One transplant center that performed routine testing for SARS-CoV-2 antibodies found that 42% of KTRs who tested positive were asymptomatic or had mild symptoms not requiring medical attention.⁴⁰

Treatment and Management of Immunosuppression

Decisions regarding reduction in immunosuppression in KTRs should be based on severity of illness and weighed against the risks of allograft loss. Factors to consider when reducing immunosuppressive medications include allograft function, history of rejection, and presence of donor-specific antibodies. Most centers recommend a 50% dose reduction or complete cessation of the antimetabolite in KTRs with COVID-19.^{41,42} In patients with worsening symptoms or evidence of COVID-19 pneumonia, the calcineurin inhibitor (CNI) should be reduced to target the lower limit of the therapeutic range, and an increase in corticosteroid dosing can be considered. In those requiring ventilatory support, CNI withdrawal should be considered, and corticosteroid dosing should be increased to prevent acute rejection.

Recipient and Donor Screening and Selection

Both donor and prospective KTRs should have negative SARS-CoV-2 reverse transcriptase polymerase chain reaction (RT-PCR) testing prior to transplantation. Screening with chest computed tomography may reveal signs of COVID-19-related disease preceding a positive test and may be considered.³⁸ Donors with active or recent COVID-19 should be declined or deferred. T cell-depleting induction therapy may reduce viral clearance and decrease the ability to mount an immune response to SARS-CoV-2. In highly sensitized patients who require potent immunosuppression, it may be reasonable to delay transplantation until vaccination. Though it is unknown how

long transplantation should be deferred in prospective KTRs who test positive, the American Society of Transplantation suggests waiting for resolution of symptoms and at least 2 negative SARS-CoV-2 RT-PCR tests.⁴³

RENIN-ANGIOTENSIN SYSTEM INHIBITORS IN COVID-19

The ACE2 receptor, the entry receptor for SARS-CoV-2, is upregulated by angiotensin-converting enzyme (ACE) inhibitors and angiotensin II (Ang II) receptor blockers (ARB), leading to theoretical concerns that these medications may increase susceptibility to COVID-19. However, ACE2 is a carboxypeptidase that reduces Ang II levels by cleaving Ang II to Ang (1–7) and by cleaving Ang I to Ang (1–9), which is converted by ACE to Ang (1–7), which binds the MAS receptor.⁴⁴ These effects of ACE2 are cytoprotective, and increased lung expression of ACE2 is associated with improved outcomes in lung injury,³ leading investigators to study whether ACE inhibitors and/or ARB are protective or harmful in patients with or at risk for COVID-19. At this time, there is no compelling evidence that ACE inhibitors/ARBs affect susceptibility to or severity of COVID-19, and major medical societies, including the American Society of Nephrology and the American Heart Association, have issued statements urging clinicians to continue prescribing these medications for patients who are likely to otherwise benefit from them.

ANTI-SARS-COV-2 THERAPIES AND VACCINES IN PATIENTS WITH KIDNEY DISEASE

At this time, no evidence-based recommendations can be made regarding the use of antiviral therapies in patients with kidney disease and COVID-19. Most clinical trials evaluating the use of these medications

have excluded patients with severe kidney disease and/or KTRs. Monoclonal anti-SARS-CoV-2 therapies do not require dose adjustment for kidney function. Remdesivir is not currently recommended in those with eGFR less than 30 mL/min/1.73 m², though small series have reported its use in patients with ESKD or AKI requiring dialysis.⁴⁵ The carrier molecule of remdesivir is eliminated by the kidney and may accumulate in those with eGFR less than 30 mL/min/1.73 m². One study using postmarketing data found that remdesivir was associated with sevenfold higher odds of AKI compared with other COVID-19 therapies.⁴⁶ Therefore, caution should be used in patients with reduced eGFR and in those at high risk for development of AKI. The optimal timing of hemodialysis relative to remdesivir dosing remains to be determined.

Data on vaccine efficacy for prevention of COVID-19 in patients with severe CKD are currently lacking. Small preliminary studies suggest that patients with ESKD receiving dialysis mount vigorous antibody responses after vaccination⁴⁷; however, KTRs may have weakened antibody responses.⁴⁸ Given the increased risk of COVID-19–related complications in patients with kidney disease, it is important to prioritize these patients for vaccination and to optimize approaches to maximize the efficacy of vaccination in this vulnerable population.

CONCLUSIONS

Acute and chronic kidney injury are important complications of COVID-19 and determinants of clinical outcomes in patients with COVID-19. Further research is needed to identify novel strategies to prevent and treat COVID-19 induced kidney injury and to determine optimal approaches to improve outcomes in patients with COVID-19 who have kidney disease.

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